

PERSONAL INFORMATION

Antonio Pedicini



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Gender Male | Date of birth 24 July 2001 (Age 24) | Nationality Italian

JOB APPLIED FOR

Internship

PROFESSIONAL SUMMARY

Master's student in Electronic Engineering specializing in Automation, with a Bachelor's degree in Computer Engineering. Having successfully completed all coursework and exams with an average score of 29/30, I am currently seeking a traineeship focused on UAVs, machine learning and optimal control strategies. I am eager to leverage my academic research and simulation experience in a professional engineering setting.

EDUCATION AND TRAINING

2024 - Present

Master's Degree in Electronic Engineering for Automation and Sensing - Università degli Studi del Sannio

– Automation Curriculum

2020 - 2024

Bachelor's Degree in Computer Engineering - Università degli Studi del Sannio

- Graduation mark: 110/110 *cum laude*
- Age at graduation: 23
- Graduation date: 31/10/2024
- Thesis Supervisor: Professor Francesco Picariello
- Co-Supervisor: Professor Ioan Tudosa
- Thesis subject: Fault Detection and Diagnosis (FDD) methods
- Thesis Title: "A data-driven testbench for fault detection of UAV propellers"

2015-2020

Scientific High School "IIS Galileo-Vetrone", Benevento

– Diploma mark: 95/100

PROJECTS

Bachelor's Thesis

A Data-Driven Testbench for Fault Detection of UAV Propellers

Supervisors: Prof. Francesco Picariello, Prof. Ioan Tudosa

- **System Design:** Engineered a holistic testbench capable of testing UAV systems as a whole, improving upon component-only literature standards.
- **Data Acquisition:** Developed software suites for sensor calibration and generated datasets distinguishing between healthy and faulty propeller states.
- **Machine Learning:** Processed statistical features to diagnose fault presence, position, and severity using MATLAB Diagnosis Feature Explorer (DFE); trained classification algorithms using MATLAB Classification Learner Toolbox, that yielded comparable results with existing literature methods.
- **Technology Stack:** Arduino, LabVIEW, Python, MATLAB.

Group Project Automated Measurement System for Flow Rate Sensor Calibration**Course:** Measurement for Automation & Industrial Production**Supervisor:** Prof. Sergio Rapuano**Grade:** 30/30 *cum laude*

- **Firmware Development:** Developed Arduino firmware to sample measurements from a reference digital flow rate sensor.
- **Hardware Design:** Designed the signal conditioning circuit for acquiring the analog Device Under Test (DUT) output.
- **Automation Logic:** Architected a LabVIEW solution utilizing Finite State Automata and Producer-Consumer paradigms to fully automate the calibration procedure.
- **Reporting:** Automated the generation of prototype Digital Calibration Certificates (DCC) upon procedure completion, significantly streamlining post-test documentation.

Group Project Bridgeless Boost PFC Converter: Control Strategies Comparison**Course:** Dynamics and Control of Switched Electronic Systems**Supervisor:** Prof. Luigi Iannelli**Grade:** 30/30 *cum laude*

- **System Modeling:** Analysed the analytical model of the converter through averaged and small-signal modelling techniques.
- **Control Strategies:** Evaluated three distinct control strategies for the Bridgeless Boost PFC Converter (BBC). Implemented the cascaded PI/PR, the Lyapunov-based inner control loop, and the dynamic phasors approach using an SOGI-PLL.
- **Simulation:** Conducted a comprehensive comparative evaluation of the control schemes through PLECS simulations.

Academic Project Reinforcement Learning for Chess Endgames**Course:** Learning for Dynamics and Control**Supervisor:** Prof. Carmela Bernardo**Grade:** 30/30 *cum laude*

- **Algorithm Design:** Explored Deep Reinforcement Learning to solve the King and Queen versus King endgame. Applied a Deep Q-learning approach to train the agent moving White's pieces.
- **Curriculum Learning:** Structured the training environment to incrementally increase in difficulty to overcome sparse rewards.
- **Neural Networks:** Evaluated and compared Fully Connected Networks (FCN) and Convolutional Neural Networks (CNN) for state evaluation.
- **Technology Stack:** Developed the simulation setup using MATLAB scripts, the Reinforcement Learning Toolbox, and tested the trained agent against the Stockfish 18 chess engine.

Academic Project Computational Thermoelectric Generator Model Study & Optimization**Course:** Multiphysics Modelling**Supervisor:** Prof. G. Castaldi**Grade:** 30/30

- **Multiphysics Simulation:** Developed a computational model of a polymer-based Thermoelectric Generator (TEG) using **COMSOL Multiphysics** to evaluate performance under thermal conditions typical of aircraft flight scenarios.
- **System Modeling:** Coupled Heat Transfer in Solids and Electrical Circuit physics interfaces to analyze the temperature distribution and electric potential in both open and closed-circuit configurations.
- **Material Engineering:** Re-designed the base model into a Segmented TEG (STEG) incorporating Skutterudite (a temperature-dependent material), demonstrating an increase in total output voltage.
- **Numerical Optimization:** Applied bound-constrained optimization algorithms (Nelder-Mead, BOBYQA, Coordinate Search) to maximize the thermal conversion efficiency (Figure of Merit) and internal resistance of the TEG module.

FOREIGN LANGUAGE SKILLS

Mother tongue Italian

Other languages

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken interaction	Spoken production	
English	B2	B2	B2	B2	B2

Levels: A1 and A2: Basic user – B1 and B2: Independent user – C1 and C2: Proficient user
[Common European Framework of Reference for Languages](#)

PERSONAL SKILLS

Communication skills Team-oriented problem solver who values clear, open communication. I naturally enjoy collaborating with others, building constructive relationships to tackle complex projects effectively, and ensuring everyone is on the same page during complex projects.

Organisational / managerial skills Skilled in task delegation, team motivation, and strategic planning. I excel at prioritizing tasks to balance demanding academic workloads while consistently delivering high-quality results on strict deadlines.

INFORMATION TECHNOLOGY SKILLS

Programming Python, C/C++, Java

Engineering Software LabVIEW, MATLAB/Simulink, PLECS, COMSOL Multiphysics

Hardware & Embedded FPGA Programming, Hardware Description Languages (Verilog), Data Acquisition (DAQ)

OTHER SKILLS

Driving licence B

Date 05 /2026

"I declare that I have read and understood the information for the processing of candidate data pursuant to EU Reg. 2016/679 and I give my consent for the processing of personal data belonging to particular categories, possibly contained in my curriculum vitae".